

EXERCISE 3: AGILE PROJECT SETUP

Aim

To set up an Agile project environment by creating user stories, epics, story mapping, and backlog prioritization for the Credit Card Processing System.

Tools Used

- Trello (Agile Board Management)
- GitHub (Version Control)
- VS Code (Development Environment)

Procedure

38. Agile Board Setup

An Agile board was created using Trello to manage and track project progress. The board consists of the following columns:

- Backlog
- To Do
- In Progress
- Testing
- Done

This structure helps in visualizing task flow and improving team productivity.

39. Swimlanes Configuration

Swimlanes were configured to categorize tasks into different work types:

- User Features
- Bug Fixes
- Technical Tasks
- Improvements

This classification helps in better organization and prioritization of tasks.

40. User Story Creation

User stories were created using the INVEST principle (Independent, Negotiable, Valuable, Estimable, Small, Testable).

User Stories:

1. As a user, I want to register an account so that I can access the system.
2. As a user, I want to log in securely so that my data is protected.
3. As a user, I want to add credit card details so that I can make payments.
4. As a user, I want to perform transactions so that I can pay bills.

5. As a user, I want to view transaction history so that I can track my expenses.
6. As a user, I want to receive payment confirmation notifications.
7. As a user, I want OTP verification for secure transactions.
8. As a system, I want to validate card details before processing payments.
9. As a system, I want to encrypt user data to ensure security.
10. As a system, I want to detect fraudulent transactions.
11. As an admin, I want to monitor all transactions.
12. As an admin, I want to block suspicious accounts.
13. As an admin, I want to generate reports.
14. As a user, I want to update my profile information.
15. As a user, I want to delete my account.
16. As a system, I want to log all activities.
17. As a user, I want fast transaction processing.
18. As a system, I want to send alerts for suspicious activity.
19. As a user, I want customer support assistance.
20. As an admin, I want dashboard analytics for insights.

41. Epic Creation

User stories were grouped into the following Epics:

- User Management
- Payment Processing
- Security & Fraud Detection
- Admin Dashboard
- Notifications & Alerts

42. Epic Breakdown

Each epic was further divided into smaller user stories.

Example:

Epic: Payment Processing

- Add Card
- Validate Card
- Process Payment
- Transaction History

This breakdown helps in better sprint planning and development.

43. Story Mapping Workshop

A story mapping activity was conducted to organize user activities in sequence:

Login → Add Card → Make Payment → Confirmation → View History

This helps in understanding the overall workflow and user journey.

44. Story Map Structure (MVP)

The Minimum Viable Product (MVP) includes the following features:

- User Registration/Login
- Add Card
- Make Payment
- Transaction History
- Basic Security (OTP Verification)

Figure 3.1: Story Map of Credit Card Processing System



Description: This diagram represents the sequential flow of user interactions in the system.

45. Definition of Ready (DoR)

A user story is considered ready when:

- Clearly defined
- Acceptance criteria are available
- Dependencies are identified
- Estimated by the team
- Testable

46. Definition of Done (DoD)

A user story is considered done when:

- Code is implemented
- Unit testing is completed
- Code is reviewed
- Integrated into main branch
- Documentation is updated

47. Backlog Prioritization

The product backlog was prioritized based on business value and urgency.

Priority	Features
High	Login, Payment Processing, OTP

Priority	Features
Medium	Transaction History, Notifications
Low	Analytics Dashboard, Reports



Azure Board Workflow

Description: This diagram shows the workflow of tasks in the Agile board.

Result

Successfully set up an Agile project environment including Agile board, user stories, epics, story mapping, and backlog prioritization for the Credit Card Processing System.